

Occupational Segregation by Race and Tipping Dynamics*

Jeongmin Ha[†]

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Abstract

This paper empirically examines whether Black-White and Hispanic-White occupational segregation exhibits tipping behavior, where White workers exit an occupation once the minority employment share crosses a “tipping point”. Using Census and American Community Survey data from 1980 to 2020, I estimate candidate tipping points with the fixed-point procedure of Card et al. (2008). Only one of eight group-by-decade combinations produces a discontinuity. The 1980 Hispanic candidate tipping points average 7.1 percent. Just above the points, non-Hispanic White employment growth over the following decade is lower by 10.4 percentage points of base-year employment. Black candidate points in every decade, and Hispanic points in later decades, give smaller estimates that are not consistent across sample splits. This suggests that tipping is not a general feature of racial and ethnic occupational segregation.

*This is a working draft. Please do not circulate.

[†]Department of Economics, Michigan State University.

1 Introduction

Over recent decades, the U.S. workforce has become more racially and ethnically diverse as employment among both Black and Hispanic workers has grown. Yet racial and ethnic occupational segregation persists (Alonso-Villar et al. 2012; Del Río and Alonso-Villar 2015; Jardina et al. 2023; Spriggs and Williams 1996). Occupational segregation matters because the allocation of workers across occupations shapes racial wage inequality and aggregate productivity (Hsieh et al. 2019; Hurst et al. 2024; Jardina et al. 2023). Existing research documents the magnitude and evolution of occupational segregation but it is less well known how White employment responds as Black or Hispanic representation in an occupation rises. This paper tests whether that response is consistent with the prediction from Schelling (1971), and exhibits tipping dynamics—White employment growth falling discontinuously once the Black or Hispanic employment share crosses a candidate threshold, tipping point.

Card et al. (2008) empirically test whether tipping patterns exist in residential segregation between race and ethnic groups. Later studies have asked whether tipping also appears in other settings, including occupational segregation by gender and workplace segregation across nativity (Pan 2015; Willis 2023). The results differ across settings. I therefore examine whether tipping appears in racial and ethnic occupational segregation.

Estimating a threshold is difficult because it is not observed and must be estimated from the same outcome used to test for a break. Searching over candidate cutoffs and reporting the best one produces large discontinuities even when none exists (Hansen 2000). I therefore separate the sample and follow the two steps procedure proposed by Card et al. (2008). Within each of the division-by-occupational groups, two-thirds of the occupation-state cells locate the candidate point and the remaining third estimates the discontinuity. Specifically, the first step looks for the minority share at which White employment in an occupation switches from growing faster than similar occupations to growing more slowly. The second step then tests whether White employment growth drops discontinuously at that candidate point. Because an occupation sample size is smaller than the neighborhood sample in the residential literature, the result might depend on the initial partition of the sample, so I redraw the partition 500 times and report how much the answer moves.

Using Census and American Community Survey data from 1980 to 2020, only one of the eight group-by-decade combinations shows a tipping pattern, the 1980–1990 Hispanic case. At the 1980 Hispanic candidate points, which average 7.1 percent, non-Hispanic White employment growth over the following decade is 10.4 percentage points of base-year cell employment lower just above the threshold. The estimate is negative in all 500 partitions. Hispanic employment grew 65.7 percent over that decade, the largest increase over the sample period, so the one break appears where the inflow was largest. The seven remaining combinations produce small estimates that are not consistent across

sample splits.

The next section presents the tipping framework. Section 3 describes the data and the split-sample estimator. Section 4 reports the results. Section 5 concludes.

2 Theoretical Framework

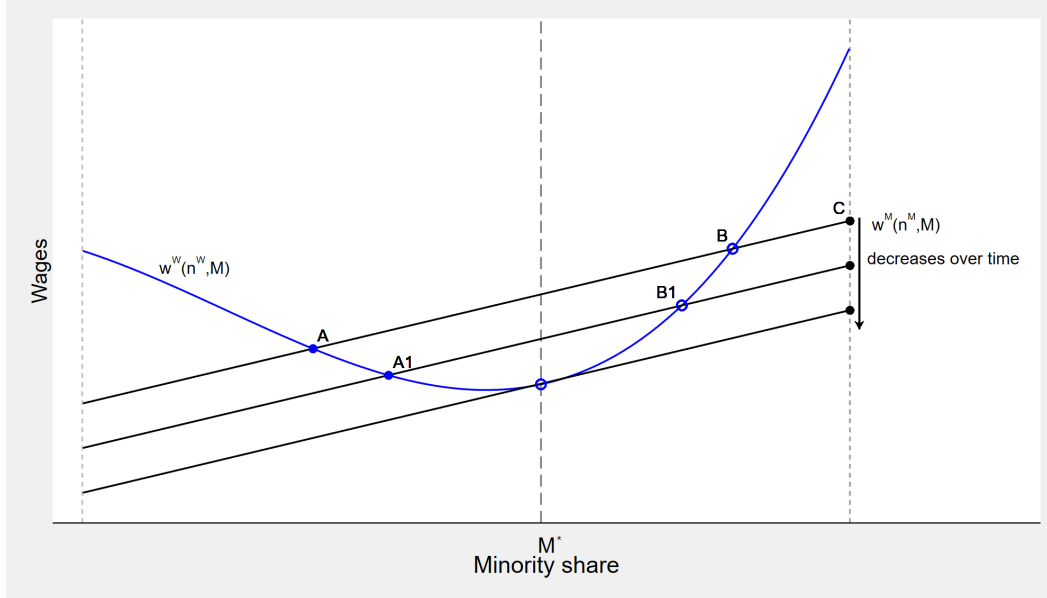
In this section I briefly present a modified version of the segregation model developed by Card et al. (2008) and Pan (2015), which study the neighborhood segregation and gender occupational segregation. Since the focus of this model is on the supply side of labor, I assume fixed demand and nondiscriminating employers hiring two types of workers, minority and White workers. Let n^W and n^M be the number of White and minority workers employed in an occupation. The labor supply of White and minority workers to an occupation depends on the wage and the minority employment share, $M = n^M / (n^W + n^M)$, that is, for an employer to hire a given quantity n^i of type i , the employer needs to pay a wage $w^i(n^i, M)$, for $i \in \{W, M\}$. I also assume that the partial derivatives of the inverse labor supply functions of White and minority workers, $\partial w^i(n^i, M) / \partial n^i$, are weakly positive. This assumption implies that, given the minority share in an occupation, the employer needs to raise wages to hire more workers of a given type.

Similar to Card et al. (2008), I assume that the social interaction effect, $\partial w^W(n^W, M) / \partial(M)$, is positive for minority shares beyond some threshold, implying that when the minority share in an occupation exceeds a critical minority share, White workers require a premium to work in the occupation. I obtain the following derivative of the inverse labor supply function of White workers under the normalization that the total labor force is one (i.e., $n^W = 1 - n^M$):

$$\frac{\partial w^W(1 - n^M, M)}{\partial(M)} = -\frac{\partial w^W(n^W, M)}{\partial(n^W)} + \frac{\partial w^W(n^W, M)}{\partial(M)}$$

Under the previous assumptions, the first term in this expression is negative, while the second term is positive when M is beyond some threshold. If the social interaction effect $\partial w^W / \partial M$ is small at $M = 0$ but dominates at high levels of M , the inverse labor supply function of White workers will have the shape plotted in Figure 1, which slopes downward in the minority share at low levels of M and upward as M rises (Card et al. 2008).

Figure 1: Theoretical effect of an increase in minority labor supply



Notes: The curved line is White inverse labor supply. The three straight lines represent successive outward shifts in minority labor supply. A and A1 are stable mixed equilibria, B and B1 are unstable mixed equilibria, and C is the all-minority equilibrium. The dashed vertical line marks the candidate tipping point.

Several models interpret the positive social interaction effect $\partial w^W(n^W, M)/\partial M > 0$. The tradition of social interaction models proposed by Schelling (1971) demonstrates that even if both groups prefer integration, they may wish to avoid minority status. Because this mechanism operates through where the composition is heading rather than through its current level, it fits most naturally where one group's share is changing quickly, as with gender segregation in Pan (2015) or an occupation absorbing a large inflow of minority. The interpretation from the model by Card et al. (2008) is that the social interaction effect is based on distaste, resulting in White worker's disutility from working with minority workers. However, as I discuss in the next section, the unit of analysis is an occupation-state cell, and it is plausible that workers of each type in an occupation-state cell do not actually interact with each other. Hence, the aversion of White workers to associating with minority workers is not entirely explained by these models. Alternatively, the disutility could arise from the presence of uncertainty about technology shocks, based on the "Pollution" model of discrimination proposed by Goldin (2014). If White workers have imperfect information about technology shocks and infer the shocks from observables, such as the racial composition of an occupation, the entry of minority workers may convey information that the occupation has undergone a negative productivity shock, even when it has not. Of the three, the Pollution channel is the only one that does not require workers of each type to meet, making it the preferred interpretation in this setting.

The effect of increasing labor supply of minority workers is depicted in Figure 1. Under the

assumption of fixed labor demand and nondiscriminating employer, the market-clearing wage is determined at the point where the wage for White and minority workers is identical. For this occupation in the figure, there are two mixed equilibria (Points A and B) and an all-minority equilibrium (Point C). While equilibria A and C are stable, equilibrium B is unstable since, after a small increase in the minority share from M , the wage demanded by White workers is higher than that demanded by minority workers, and minority workers in turn displace more expensive White workers until the occupation reaches equilibrium C.

If there is an exogenous increase in the labor supply of minority workers over time, the inverse labor supply function of minority workers $w^M(n^M, M)$ shifts downward, resulting in a new equilibrium at A1. As the supply of minority workers increases further, the equilibrium eventually reaches the point where w^M is just tangent to w^W . At this point of tangency, equilibrium is still stable with decreases in minority share, but unstable with respect to increases. That is, any further increase in the supply of minority workers leads to a fully segregated equilibrium ($M = 1$). The minority share at this tangency point is a “tipping point.”

In this model, I assumed steady increases in the labor supply of minority workers. Since there has been an increase in Black and Hispanic labor supply in the US labor market over time, this assumption is likely to be true, on average. When each occupation faces an increasing labor supply of minority workers, the framework shows several empirical implications. First, if occupations begin with an initial minority share well below the tipping point, a small shift in the relative labor supply of minority workers moves them toward a new integrated equilibrium, as long as the minority share remains below the tipping point. Second, occupations that already have a minority share above the tipping point move toward the fully segregated equilibrium ($M = 1$). Last, occupations with an initial minority share just below the tipping point will tip when they face a shift in minority labor supply, showing a discontinuous and negative change in the White share. Based on these implications, I expect a discontinuous change in racial composition for occupations whose minority share passes just beyond the tipping point, relative to occupations just below it.

3 Data and Empirical Strategy

3.1 Data and Group Definitions

The base years are 1980, 1990, 2000, and 2010, and each is paired with the endpoint ten years later. The 1980, 1990, and 2000 endpoints use the five-percent Census samples from the Integrated Public Use Microdata Series (IPUMS). The 2010 endpoint uses the 2008–2012 American Community Survey five-year sample and the 2020 endpoint uses the 2018–2022 sample, each assigned to its midpoint year. Pooled ACS endpoints are period averages rather than single-year observations.

The sample covers employed workers age 16 and older. Occupation is the harmonized IPUMS OCC1990 variable at every endpoint, mapped to the detailed occupation classification of Autor and Dorn (2013) and Deming (2017). I exclude agricultural occupations, which cover farming, forestry, and fishing, along with military occupations and observations with no classifiable occupation code. The four analysis panels contain 318 occupation codes in the first two transitions, 315 in the third, and 314 in the fourth, so the classification is stable across the period the paper studies.

The unit of observation is an occupation-state cell. Adding states introduces regional variation while leaving enough cells inside each search group to locate a threshold. Cells are included only if they contain at least 30 individual observations at both endpoints. Table 1 shows that this leaves between 9,952 and 10,763 cells per transition, with a median of 146 to 195 observations and 2,920 to 4,027 workers in the base year.

Table 1: Descriptive statistics for the analysis panels

	1980–1990	1990–2000	2000–2010	2010–2020
Occupation–state cells	9,952	10,302	10,763	10,684
Median observations per base-year cell	146	163	185	195
Median base-year employment	2,920	3,333	3,830	4,027
Mean age	37.5 (3.7)	38.1 (3.4)	39.6 (3.7)	41.6 (4.2)
Mean years of schooling	12.5 (1.9)	13.0 (1.7)	13.2 (1.7)	13.5 (1.8)
Female share (%)	43.7 (32.3)	46.4 (29.9)	47.3 (29.4)	48.2 (28.8)
Part-time share (%)	18.4 (15.1)	18.8 (14.4)	18.8 (14.7)	22.0 (16.5)
NH White employment share (%)	82.7 (14.4)	79.0 (15.9)	73.8 (17.2)	67.1 (19.5)
Black employment share (%)	9.7 (10.5)	10.0 (9.9)	10.2 (9.4)	11.0 (10.1)
Hispanic employment share (%)	5.5 (8.1)	7.7 (10.6)	10.2 (12.6)	14.8 (15.9)
Black-or-Hispanic employment share (%)	15.1 (13.3)	17.5 (14.2)	20.2 (15.2)	25.5 (18.0)
Total employment growth (percentage points)	19.2 (44.5)	13.1 (107.8)	9.8 (26.0)	12.2 (28.5)
Net NH White-minus-Black growth (percentage points)	8.7 (29.3)	-1.1 (28.5)	-2.0 (15.7)	-1.6 (16.3)
Net NH White-minus-Hispanic growth (percentage points)	7.3 (31.1)	-3.4 (30.1)	-6.2 (17.9)	-4.7 (16.9)

Notes: Each column describes occupation–state cells with at least 30 observations at both endpoints. Base-year characteristics, group shares, and growth outcomes are weighted by base-year PERWT-expanded employment; standard deviations are in parentheses. Cell counts and medians are unweighted. NH White denotes non-Hispanic White. Black and Hispanic categories overlap; Black-or-Hispanic is their union. Growth outcomes divide the employment change by base-year total employment. The 2010 and 2020 endpoints are the 2008–2012 and 2018–2022 ACS five-year samples.

The analysis uses two comparisons. The first pairs non-Hispanic White workers with Black workers and the second pairs them with Hispanic workers. A non-Hispanic White worker reports White race and no Hispanic origin. A Black worker reports Black race whatever their origin, and a Hispanic worker reports Hispanic origin whatever their race, so the two minority groups overlap. I refer to non-Hispanic White workers as White workers from here on.

The racial composition of these cells shifted over the period. Table 1 shows the employment-weighted White share falling from 82.7 to 67.1 percent across the four base years. The Black share

changes from 9.7 to 11.0 percent while the Hispanic share nearly triples, from 5.5 to 14.8 percent. Table 2 reports the corresponding growth rates. Hispanic employment grew 65.7 percent from 1980 to 1990 against 23.2 percent for Black employment and 14.2 percent for White employment, and it grew faster than either group in every subsequent transition.

Table 2: Aggregate employment growth by race and ethnicity

Period	Total	NH White	Black	Hispanic	Black or Hispanic
1980-1990	19.5	14.2	23.2	65.7	38.4
1990-2000	12.7	5.4	14.7	49.3	30.1
2000-2010	9.6	-0.3	18.6	59.3	38.7
2010-2020	11.9	0.8	19.7	35.9	29.1

Notes: Entries are percentage changes in employment within the sample used in this paper. They use all eligible nonagricultural occupation-state observations and do not impose the cell-size restriction. The 1980, 1990, and 2000 endpoints use five-percent Census samples. The 2010 and 2020 endpoints use the 2008–2012 and 2018–2022 ACS five-year samples. The ACS endpoints are centered period averages rather than point-in-time estimates. The IPUMS person’s weight is used. NH White denotes non-Hispanic White. Black and Hispanic categories overlap. Agricultural, military, and unclassifiable occupations are excluded.

Candidate points are allowed to vary by census division, by broad occupational group (blue- and white-collar), and by decade. The eighteen division-by-collar groups allow for geographic and occupational differences while leaving enough cells in each group to locate a threshold. Appendix Table A1 reports the cells available inside them.

3.2 Outcome and Two-step Procedure

Let $G \in \{B, H, BH\}$ denote Black, Hispanic, or Black-or-Hispanic workers. For occupation i , state s , census division r , collar group j , and end year t , White employment growth is

$$D_{isrj,t}^W = \frac{W_{isrj,t} - W_{isrj,t-10}}{P_{isrj,t-10}},$$

where W is White employment and P is total employment. Parallel measures are constructed for each minority group and for total employment, and every change is divided by base-year total employment.

The running variable, which places a cell relative to the threshold, is the base-year employment share of the relevant minority group,

$$m_{isrj,t-10}^G = \frac{G_{isrj,t-10}}{P_{isrj,t-10}}.$$

I apply the two-step procedure to test whether White employment growth fall discontinuously around a threshold. The first step searches for a candidate tipping point, and the second step finds discontinuity around the candidate tipping point. Specifically, in the first step, the search for a candidate point follows the fixed point approach in Card et al. (2008). The tipping point in the model in Section 2 is an unstable equilibrium, so composition is stationary there. Below it an occupation draws White workers faster than comparable occupations, and its minority share falls. Above the tipping point, White workers arrive more slowly and the minority share rises. Relative White employment growth therefore crosses zero from above at the tipping point, and the search looks for that crossing.¹

The second step estimates

$$D_{isrj,t}^W = p(\delta_{isrj,t-10}^G) + d^G \mathbb{1}[\delta_{isrj,t-10}^G > 0] + \tau_{rj} + \epsilon_{isrj,t}, \quad (1)$$

where $\delta_{isrj,t-10}^G = m_{isrj,t-10}^G - m_{rj,t-10}^{G*}$, $p(\cdot)$ is a fourth-order polynomial, and τ_{rj} denotes division-by-collar fixed effects. Equation 1 is estimated separately for each transition with standard errors clustered by state. A negative d^G means that White employment growth is lower just above the candidate point.

Locating the fixed point and estimating the discontinuity on the same cells produces specification-search bias (Hansen 2000). I follow the sample allocation of Card et al. (2008). Within each division-by-collar group, exactly two-thirds of the cells go to the search sample and the remaining third goes to the estimation sample, estimating equation 1. The roles are never exchanged, so no second-step outcome enters the search. Roughly 3,000 to 3,500 cells per transition remain for estimation. To ensure the results are not driven by a single random split, I repeat this partitioning process 500 times and report the consistency of the estimates across all repetition.²

One caveat applies to every estimate below. This is not a regression discontinuity design with an externally assigned cutoff. The cutoff is estimated from the outcome data, under the assumption that a break exists somewhere. A negative d^G is evidence consistent with tipping only when the threshold

¹Formally, let $\hat{f}_{rj}$ be a fourth-order polynomial in m^G fitted to $D_{isrj,t}^W - \bar{D}_{rj}^W$, where $\bar{D}_{rj}^W$ is average White employment growth in group rj . The candidate point is a downward crossing of that polynomial,

$$m_{rj,t-10}^{G*} \in \{ m < 0.5 : \hat{f}_{rj}(m) = 0, \hat{f}'_{rj}(m) < 0 \},$$

and when several roots qualify the one with the most negative slope is taken. The polynomial is fitted twice, first to cells with $m^G < 0.6$ and then to cells within ten percentage points of the first root, and the second fit gives $m_{rj,t-10}^{G*}$. The 60 percent bound keeps the polynomial from extrapolating at the edge of its support. The 50 percent bound on the root follows from the model, in which the tipping point is where White workers begin to leave and so lies below the share at which they are already outnumbered. Neither bound is close to binding, since 0.6 percent of the estimated roots exceed 45 percent and 0.1 percent exceed 49 percent. A group is excluded if it holds fewer than 100 eligible cells or if either fit has no root below 50 percent.

²Appendix Table ?? reports the same specification without splitting. The 1980–1990 coefficient is about three times larger on the full sample than under repeated splitting, and the same pattern holds in the other three transitions.

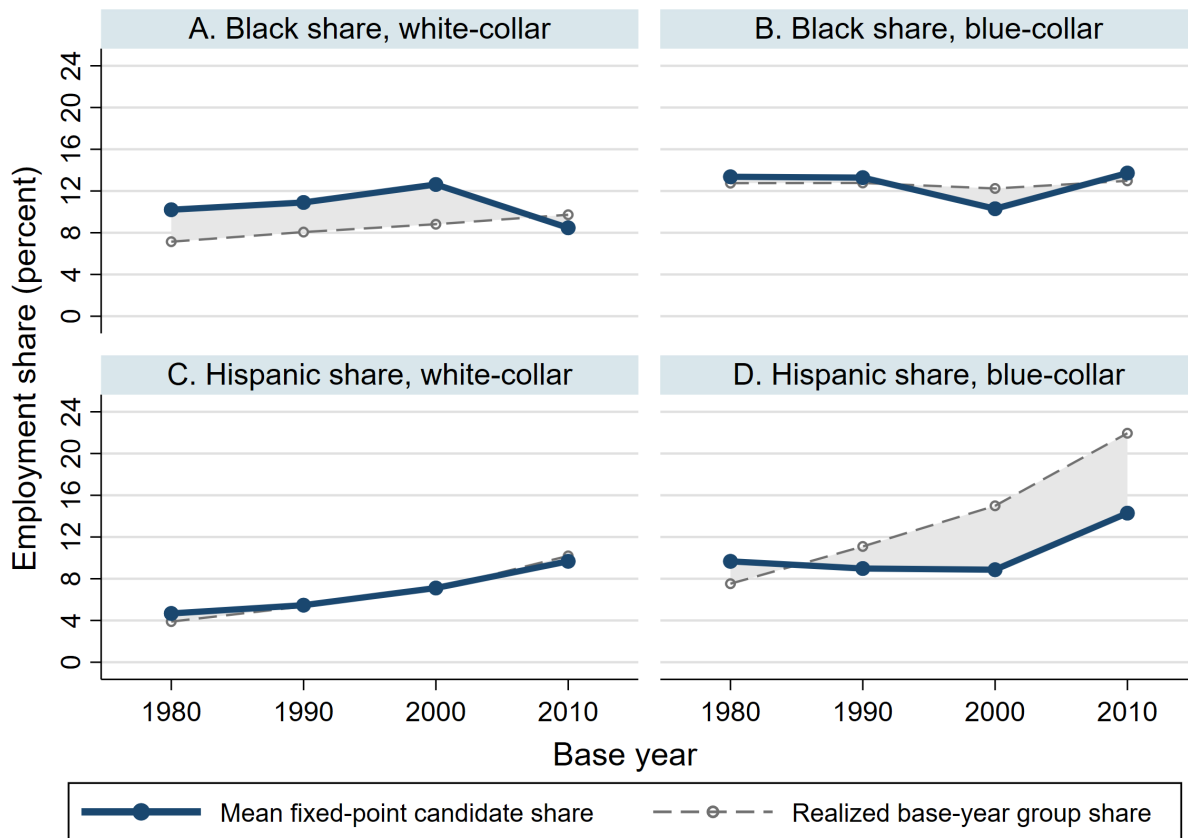
from the first step still produces a break on cells from the estimation sample in the second step.

4 Results

4.1 Candidate Tipping Points and Discontinuities

Figure 2 plots the mean candidate tipping point in each base year against the actual share of the group in the same cells, one panel per minority and collar group. The solid line indicates the estimated structural break, while the dashed line represents the initial minority employment share at the beginning of the decade. The shaded area captures the distance between the two.

Figure 2: Fixed-point candidate shares and realized group shares by race and collar, 1980–2010



Notes: Each panel is one race-by-collar combination. The solid line is the mean fixed-point candidate share, pooling division-specific roots across the 500 search samples within that collar group. The dashed line is the realized base-year employment share of the same group among the occupation–state cells that clear the thirty-observation screen at both endpoints, weighted by base-year employment. Both series are read on the same scale, so the shaded gap between them is the distance between where the search locates a candidate and where the group’s employment share actually stood. A candidate point above the realized share means most cells began the decade below the threshold.

The two Black panels exhibit no discernible trend in either series. Candidate points range from 8.5 to 13.7 percent, and the realized Black share changes little, from 7.1 to 9.7 percent in white-collar occupations and from 12.8 to 13.0 percent in blue-collar ones. In contrast, the Hispanic panels exhibit a strong upward trend. Candidate points increase from 4.7 to 9.7 percent in white-collar occupations and from 9.7 to 14.3 percent in blue-collar, tracking the actual Hispanic share that rises faster, from 3.9 to 10.2 and from 7.5 to 22.0 percent, respectively. Although these candidate locations must be verified as actual tipping points in the second step, this upward trend is consistent with Card et al. (2008) who report neighborhood tipping points. That the Black candidate points remain flat while the Hispanic thresholds rise alongside the overall Hispanic share suggests these locations track relative group size rather than a broad shift in racial preferences.

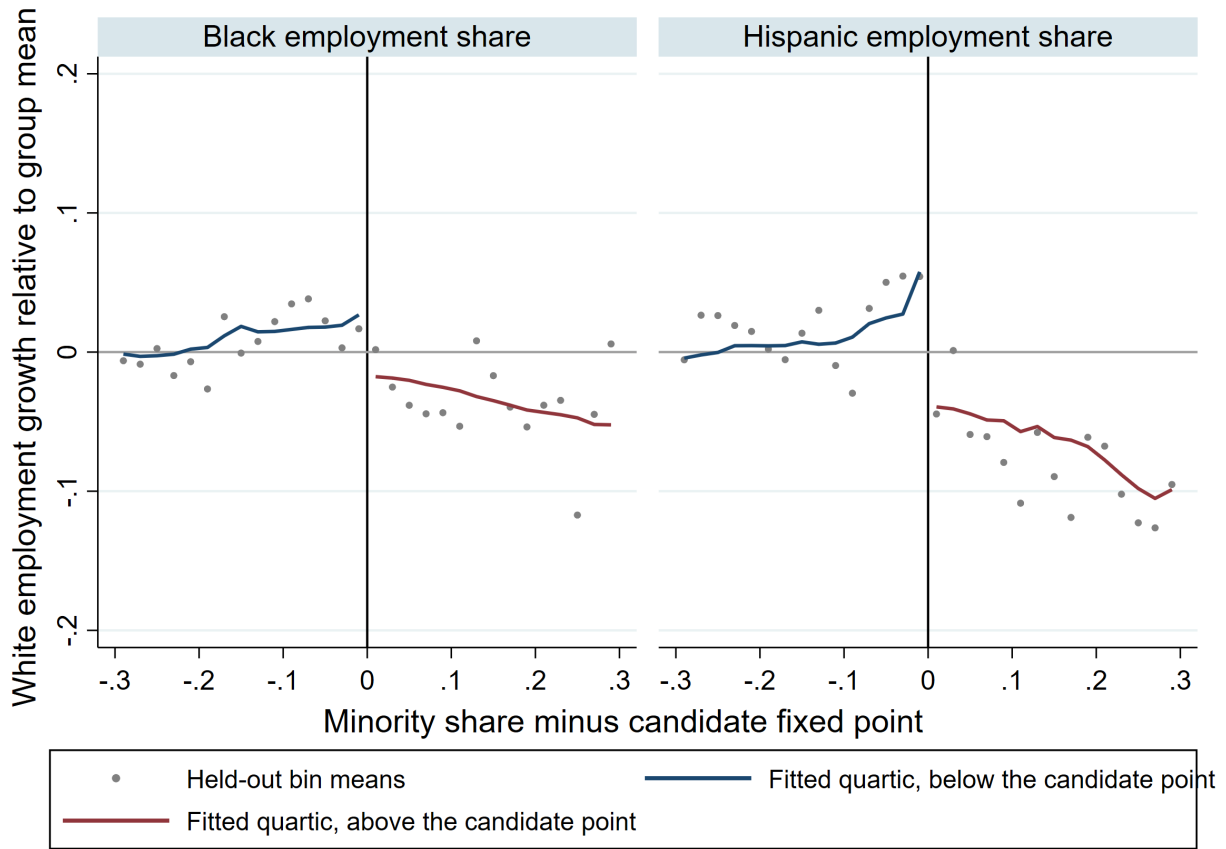
Table 3 presents the estimated discontinuities in the second-step. At the 1980 Hispanic candidate points, White employment growth between 1980 and 1990 is 10.4 percentage points lower just above the threshold, relative to base-year employment. The single split yields an estimate of -0.115 (standard error: 0.020); the mean across repeated splits is -0.104 , with a 5th-to-95th percentile range of $[-0.139, -0.068]$. This estimate is negative in all 500 partitions and statistically significant at the 5-percent level in 498 of them. To contextualize this magnitude, average cell employment grew by 19.2 percentage points over this decade, while aggregate White employment grew by 14.2 percent. Figure 3 illustrates the discontinuity in the bin means from the estimation sample for Black and Hispanic estimates, and Appendix Figure A1 and A2 plot estimates for all four transitions. In these figures, only the 1980-1990 Hispanic-White sample provides visually clear evidence of discontinuity. Appendix Figure A3 shows how the Hispanic estimate is distributed across the 500 partitions.

Table 3: Fixed-point estimates of non-Hispanic White employment growth

Period	One split estimate	Repeated-split mean	5th-95th percentile	Share negative	Share negative with $p < .05$	Mean candidate share
<i>Panel A: Black employment share</i>						
1980-1990	-0.026 (0.034)	-0.048	$[-0.102, 0.010]$	92	32	11.7
1990-2000	-0.037 (0.031)	-0.012	$[-0.058, 0.034]$	68	5	12.0
2000-2010	-0.024 (0.016)	-0.030	$[-0.055, -0.003]$	97	49	11.5
2010-2020	-0.058*** (0.020)	-0.043	$[-0.068, -0.017]$	99	73	11.1
<i>Panel B: Hispanic employment share</i>						
1980-1990	-0.115*** (0.020)	-0.104	$[-0.139, -0.068]$	100	100	7.1
1990-2000	-0.050 (0.031)	-0.050	$[-0.091, -0.007]$	97	47	7.2
2000-2010	0.014 (0.016)	-0.021	$[-0.045, 0.004]$	92	29	8.0
2010-2020	-0.019 (0.016)	-0.025	$[-0.051, 0.001]$	94	29	12.1

Notes: The outcome is the change in non-Hispanic White employment divided by base-year total employment. Candidate points are estimated separately by division and collar group with the fixed-point procedure in Card et al. (2008). Each partition assigns exactly two-thirds of the cells in every group to the search sample and the remaining third to estimation. The one-split standard error is clustered by state. Stars denote significance in that partition at the 10, 5, and 1 percent levels. The remaining columns summarize 500 independently generated partitions. Black workers include those of Hispanic origin. Hispanic origin is defined independently of race.

Figure 3: Black-share and Hispanic-share fixed points in 1980, fitted specification



Notes: Dots are two-percentage-point bin means of non-Hispanic White employment growth on estimation-sample occupation–state cells, taken net of the estimation-sample division-by-collar mean and averaged across the same 500 partitions used in estimation. The curves are the fitted values of the estimated equation itself, namely one fourth-order polynomial in distance from the candidate point common to both sides plus a level shift above zero, with division-by-collar fixed effects. They are demeaned on the same cells as the dots, so the vertical gap at zero is the reported discontinuity. Earlier drafts drew a separate straight line on each side, which is not the function that was estimated.

Estimates for subsequent decades are substantially smaller. The Hispanic discontinuity attenuates to -0.050 in the 1990s (significant in 47 percent of partitions), and further to -0.021 and -0.025 in the latter two transitions. In these recent periods, the percentile ranges cross zero.

The pattern for Black tipping points contrasts with the Hispanic results. While the Hispanic discontinuities are concentrated in the first two transitions, the Black discontinuities emerge primarily in the latter two. The 2000–2010 mean estimate is -0.030 (range: $[-0.055, -0.003]$), and the 2010–2020 mean is -0.043 (range: $[-0.068, -0.017]$). The latter is negative in 99 percent of the partitions and statistically significant in 73 percent. Although the 1980–1990 Black mean of -0.048 is similar in magnitude, its range covers zero, and only 32 percent of partitions are significant. Appendix Figure

A2 plots the estimation-sample bin means across all four Black transitions, and Appendix Figure A4 shows how the Black estimate is distributed across the same partitions.

4.2 Employment Dynamics at the Threshold

The theoretical framework in Section 2 predicts a discontinuous decline in the White share once the threshold is crossed. To investigate this, Table 4 holds the 1980 candidate points and random partitions fixed, varying only the second-step dependent variable. This ensures that every row evaluates the same set of cells at identical cutoffs.

Table 4: Decomposition of the 1980 fixed-point discontinuities

Outcome	Black share		Hispanic share	
	Mean	5th–95th percentile	Mean	5th–95th percentile
Non-Hispanic White growth	-0.048	[-0.102, 0.010]	-0.104	[-0.139, -0.068]
Stated minority-group growth	-0.008	[-0.020, 0.004]	-0.010	[-0.017, -0.004]
Total employment growth	-0.046	[-0.109, 0.019]	-0.113	[-0.153, -0.072]
White-minus-minority growth	-0.040	[-0.091, 0.016]	-0.093	[-0.128, -0.060]

Notes: Every row uses the fixed points selected from non-Hispanic White employment growth in the same two-thirds search sample. Outcomes are estimated on the estimation sample and divided by base-year total employment. The stated minority group is Black in columns 1 and 2 and Hispanic in columns 3 and 4. Percentile ranges are taken across the 500 partitions.

Around the Hispanic candidate points, in 1980-1990, the discontinuity in White employment growth is -0.104 , Hispanic employment growth is -0.010 (range: $[-0.017, -0.004]$), and total employment growth is -0.113 . Because White employment decreases proportionally faster than total employment, the White share declines above the threshold, consistent with the model's prediction. Appendix Table A3 shows the decomposition results in all four transitions.

The decomposition for the Black sample is less conclusive, as none of the component estimates are statistically distinguishable from zero. The discontinuity in White employment growth is -0.048 (range: $[-0.102, 0.010]$), compared to -0.008 for Black employment growth and -0.046 for total employment.

The 1980 Black candidate points are not where the Black evidence is, however. Appendix Table A4 applies the same decomposition to the two transitions in which the Black discontinuity is stable. The question there is whether total employment falls alongside White employment, as it does at the 1980 Hispanic points, or whether the decline is specific to White workers. The second pattern would be closer to the tipping mechanism than anything else in this paper.

4.3 Robustness and Alternative Interpretations

Three robustness exercises evaluate whether the 1980 Hispanic result is sensitive to the definitions of the outcome variable and the minority group. First, substituting net White-minus-Hispanic employment growth for White employment growth yields a discontinuity of -0.093 (range: $[-0.128, -0.060]$), as shown in Appendix Table A2. The discontinuity is negative in every partition and significant in 99 percent of partitions. Second, replacing the Hispanic share with the union of Black and Hispanic workers does not reproduce the discontinuity. The 1980 mean falls to -0.053 and only 36 percent of partitions are significant, against 100 percent under the Hispanic share. The discontinuity follows the Hispanic share rather than the minority share in general. This attenuation is consistent with established patterns of occupational segregation (Hellerstein and Neumark 2008). Because Black and Hispanic workers historically sort into distinct segments of the labor market, aggregating them into a single minority category mutes the specific demographic shifts that trigger localized tipping dynamics.

Lastly, I also located thresholds with a structural-break method (Card et al. 2008), which looks directly for the share at which the outcome falls most sharply in the first step. That design screens out groups whose first-step break is imprecise, so fewer division-by-collar groups contribute cells to the second step and the estimates are correspondingly less precise. Its largest and most consistent estimate is again Hispanic 1980–1990. At the Black candidate points and in the later Hispanic periods it gives smaller estimates with ranges that cover zero, as the fixed-point procedure does (Appendix Table A5). The details of this method is discussed in A.2.

The empirical design cannot directly explain why the structural break is specific to the 1980s. Hispanic employment grew by 65.7 percent between 1980 and 1990—the largest increase reported in Table 2—and Figure 2 places that decade’s candidate points above the prevailing Hispanic share. Consequently, many cells began the decade below the threshold during a period of rapid Hispanic entry. However, demographic expansion alone is insufficient, as Hispanic employment grew by a comparable 59.3 percent from 2000 to 2010 yet produced a much smaller discontinuity of -0.021 . The 1980s were also characterized by significant industrial and regional demand shifts that reallocated employment unevenly across occupations (Bound and Freeman 1992). While the available data do not allow for a clean separation between endogenous compositional responses and these broader occupational demand shocks, the negative discontinuity in total employment provides suggestive evidence that these unique structural shifts played a contributing role.

5 Conclusion

This paper tests whether racial and ethnic occupational segregation exhibits tipping dynamics analogous to those in residential neighborhoods. By applying the fixed-point procedure of Card et al. (2008) to occupation-state cells from 1980 to 2020, the empirical design strictly separates the search for candidate thresholds from the estimation of employment discontinuities. Repeating the random sample partition ensures that the estimates do not rely on an arbitrary sample split.

The evidence indicates that tipping is not a generalized phenomenon in the occupational context. A pronounced structural break is isolated to a single decade and demographic group. At the 1980 Hispanic candidate points, White employment growth over the subsequent decade is 10.4 percentage points lower just above the threshold, an estimate that remains negative across all 500 partitions. However, this discontinuity attenuates in each subsequent transition. Conversely, discontinuities for Black workers are substantially smaller and lack consistency, with only suggestive evidence emerging in the 2010–2020 period. Out of eight group-by-decade transitions evaluated, only one yields a large, robust discontinuity.

The concentration of the tipping dynamic in the 1980–1990 Hispanic sample highlights the episodic nature of this phenomenon. This specific decade combined a rapid demographic influx—Hispanic employment grew by 65.7 percent—with severe industrial and regional reallocations (Bound and Freeman 1992). While the empirical design filters out smooth macroeconomic trends to isolate the sharp behavioral response at the threshold, the broader structural turbulence of the 1980s likely created the unique conditions necessary for this tipping dynamic to manifest.

Ultimately, the lack of consistent structural breaks across other demographic groups and time periods demonstrates that tipping is not a universal mechanism of occupational segregation. To better understand these isolated episodes, future research would benefit from worker flow data to distinguish whether the relative decline in White employment is driven by the active exit of incumbents or a reduction in new entry.

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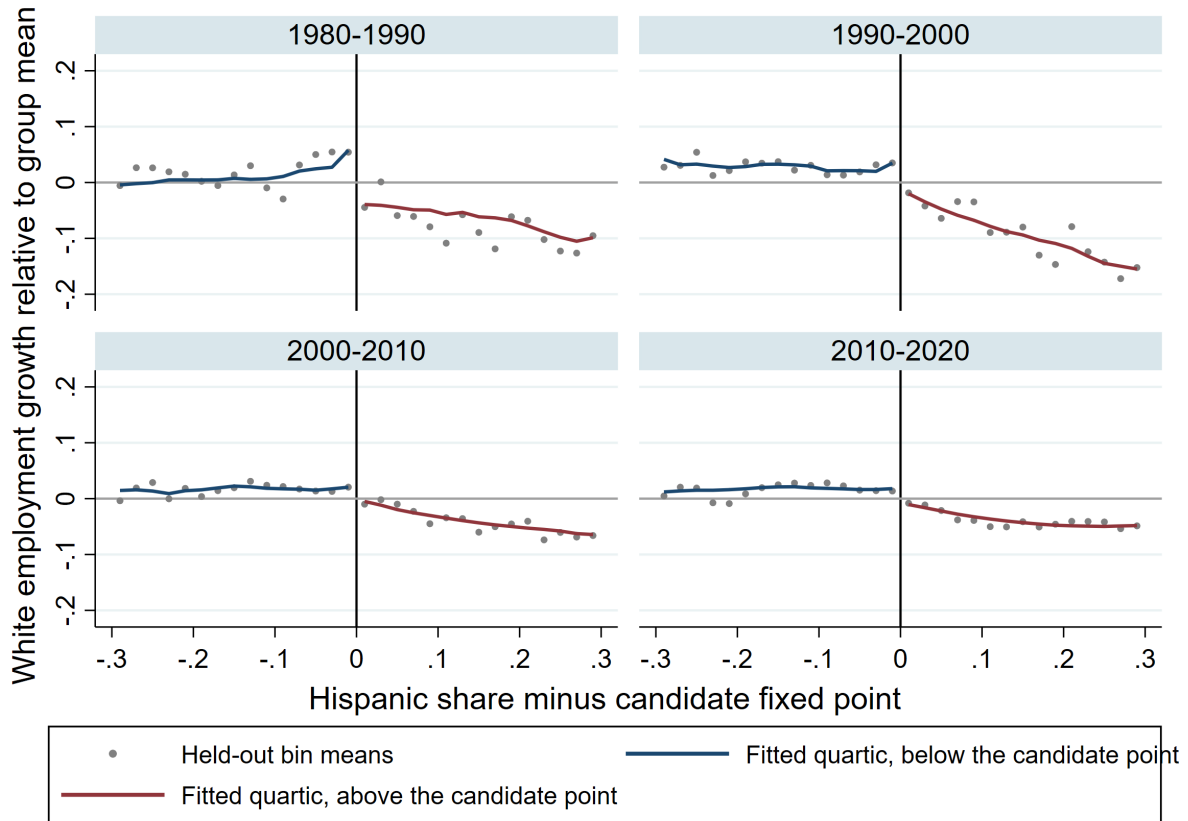
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A Supplementary Figures and Tables

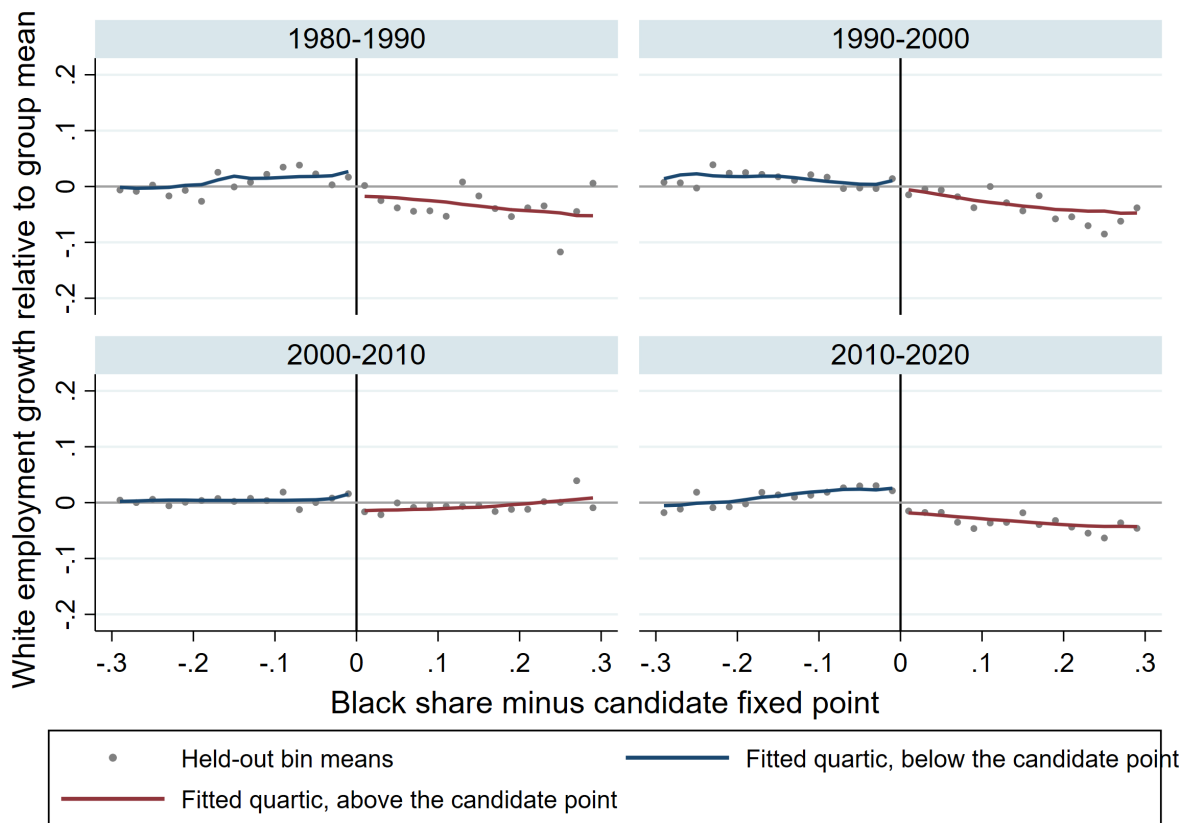
A.1 Additional Fixed-Point Results

Figure A1: Non-Hispanic White employment growth around Hispanic fixed points, fitted specification



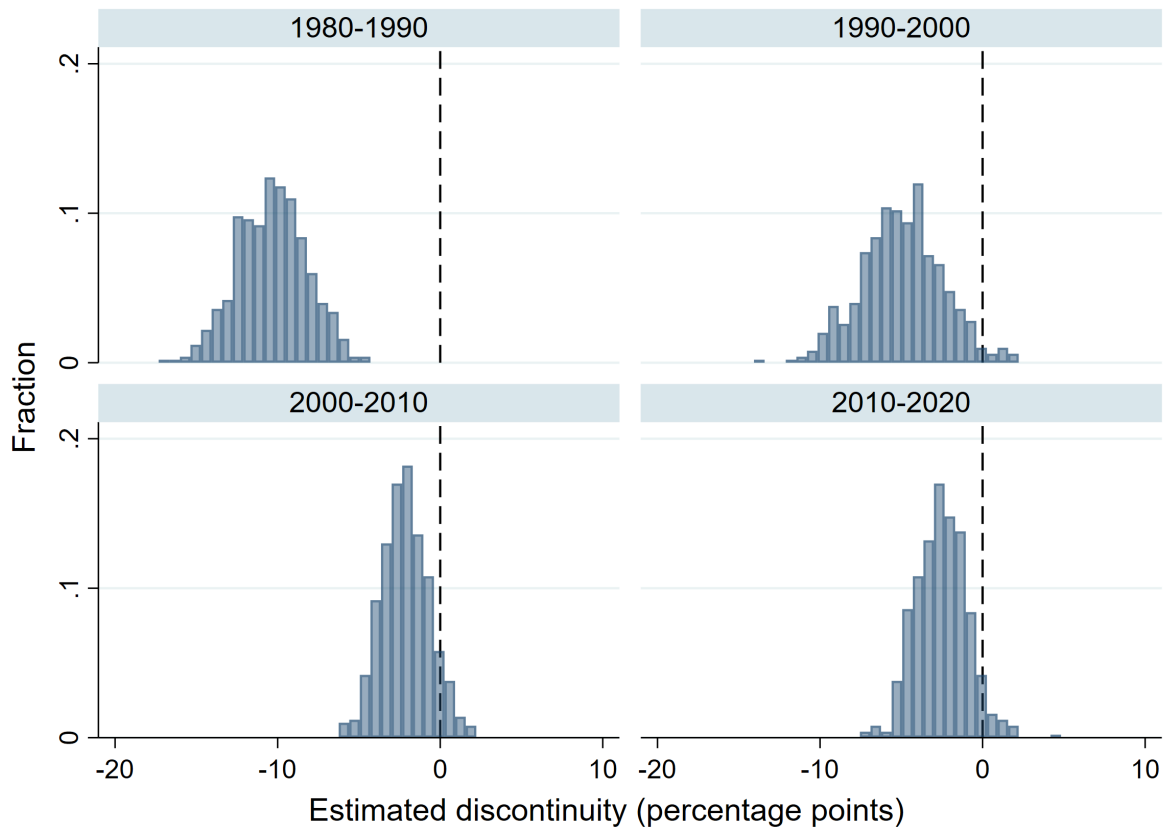
Notes: Dots are two-percentage-point bin means of non-Hispanic White employment growth on estimation-sample occupation-state cells, taken net of the estimation-sample division-by-collar mean and averaged across the same 500 partitions used in estimation. The curves are the fitted values of the estimated equation itself, namely one fourth-order polynomial in distance from the candidate point common to both sides plus a level shift above zero, with division-by-collar fixed effects. They are demeaned on the same cells as the dots, so the vertical gap at zero is the reported discontinuity. Earlier drafts drew a separate straight line on each side, which is not the function that was estimated.

Figure A2: Non-Hispanic White employment growth around Black fixed points, fitted specification



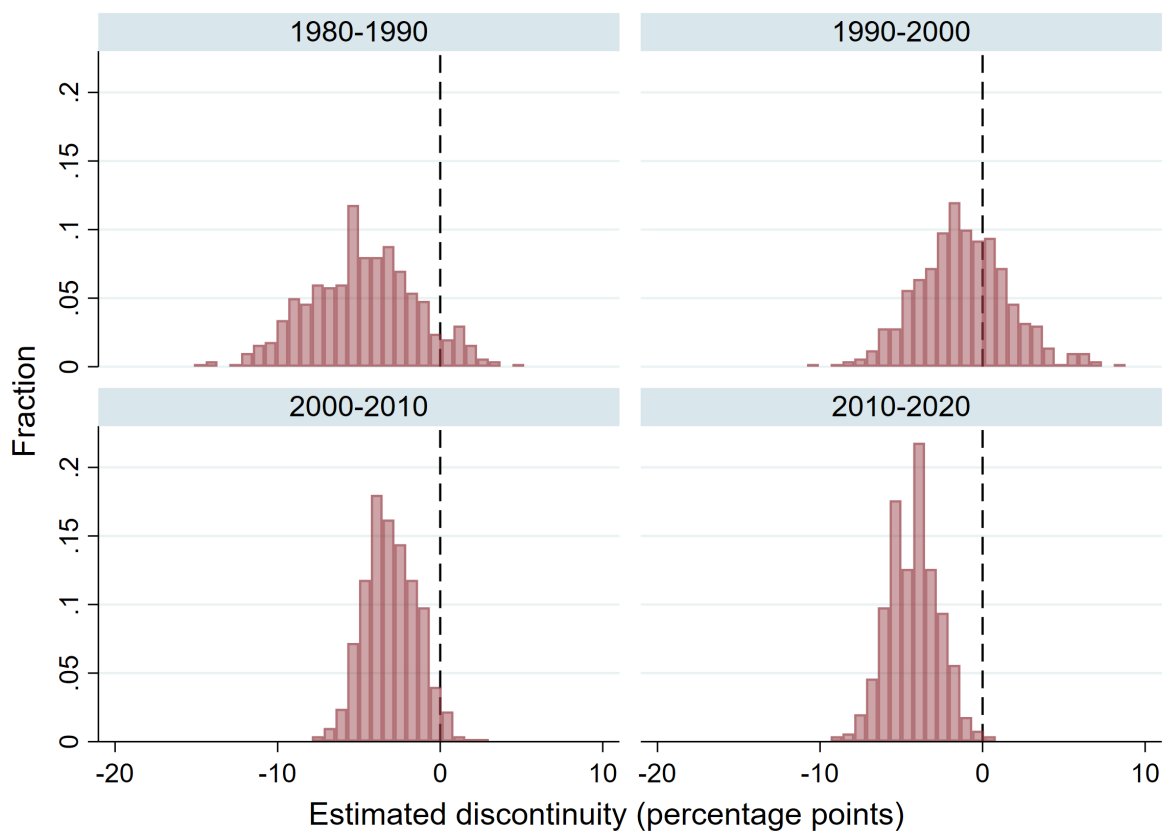
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Figure A3: Distribution of Hispanic-share fixed-point discontinuity estimates



Notes: Each panel shows the estimates from the completed exact within-group partitions. The dashed line marks zero. The distributions measure sensitivity to the random split and are not sampling distributions.

Figure A4: Distribution of Black-share fixed-point discontinuity estimates



Notes: Construction follows Figure A3, using the Black employment share as the running variable.

Table A1: Sample size available to the structural-break search, by decade

	Occupation	Occupation-	Usable	Search	Obs. per group	
	codes	state cells	cells	sample	Median	Minimum
1980-1990	318	16092	9856	5555	272	192
1990-2000	318	16073	10051	5610	282	184
2000-2010	315	15916	10761	6073	308	192
2010-2020	314	15781	10662	5998	300	190

Notes: Usable cells are occupation-state cells with at least 30 observations at both endpoints and a non-missing outcome. The search sample restricts these to one representative two-thirds search sample with a base-year Black share in $(0, 0.5]$. The last two columns describe the observations available inside each of the eighteen division $\times$ collar groups within which the candidate tipping point is located. The analysis begins in 1980 so every endpoint uses a five-percent Census or ACS sample with detailed harmonized occupations. The 2010 and 2020 endpoints are the 2008–2012 and 2018–2022 ACS five-year samples.

Table A2: Fixed-point estimates using White-minus-minority employment growth

Period	Group	Repeated mean	5th–95th percentile	Share negative	Share negative, $p < .05$	Candidate share
1980-1990	Black	-0.040	[-0.091, 0.016]	89	27	11.7
1990-2000	Black	-0.009	[-0.051, 0.036]	65	5	12.0
2000-2010	Black	-0.025	[-0.050, 0.002]	94	37	11.5
2010-2020	Black	-0.038	[-0.062, -0.013]	99	64	11.1
1980-1990	Hispanic	-0.093	[-0.128, -0.060]	100	99	7.1
1990-2000	Hispanic	-0.040	[-0.081, -0.001]	95	35	7.2
2000-2010	Hispanic	-0.014	[-0.035, 0.011]	84	13	8.0
2010-2020	Hispanic	-0.019	[-0.046, 0.008]	88	19	12.1

Notes: The outcome subtracts the employment growth of the stated minority group from non-Hispanic White employment growth. Other details follow Table 3.

Table A3: Hispanic fixed-point decomposition by period

Period	Outcome	Mean	5th–95th percentile	Share negative
1980-1990	Non-Hispanic White growth	-0.104	[-0.139, -0.068]	100
1980-1990	Hispanic growth	-0.010	[-0.017, -0.004]	99
1980-1990	Total employment growth	-0.113	[-0.153, -0.072]	100
1990-2000	Non-Hispanic White growth	-0.050	[-0.091, -0.007]	97
1990-2000	Hispanic growth	-0.009	[-0.018, -0.002]	97
1990-2000	Total employment growth	-0.053	[-0.102, -0.001]	95
2000-2010	Non-Hispanic White growth	-0.021	[-0.045, 0.004]	92
2000-2010	Hispanic growth	-0.008	[-0.015, 0.001]	92
2000-2010	Total employment growth	-0.027	[-0.060, 0.007]	92
2010-2020	Non-Hispanic White growth	-0.025	[-0.051, 0.001]	94
2010-2020	Hispanic growth	-0.006	[-0.017, 0.005]	87
2010-2020	Total employment growth	-0.027	[-0.057, 0.007]	89

Notes: Candidate points are selected from non-Hispanic White employment growth. Other details follow Table 4.

Table A4: Black fixed-point decomposition by period

Period	Outcome	Mean	5th–95th percentile	Share negative
1980-1990	Non-Hispanic White growth	-0.048	[-0.102, 0.010]	92
1980-1990	Black growth	-0.008	[-0.020, 0.004]	87
1980-1990	Total employment growth	-0.046	[-0.109, 0.019]	88
1990-2000	Non-Hispanic White growth	-0.012	[-0.058, 0.034]	68
1990-2000	Black growth	-0.003	[-0.014, 0.008]	68
1990-2000	Total employment growth	0.000	[-0.059, 0.058]	51
2000-2010	Non-Hispanic White growth	-0.030	[-0.055, -0.003]	97
2000-2010	Black growth	-0.006	[-0.013, 0.002]	90
2000-2010	Total employment growth	-0.019	[-0.050, 0.011]	83
2010-2020	Non-Hispanic White growth	-0.043	[-0.068, -0.017]	99
2010-2020	Black growth	-0.004	[-0.012, 0.003]	87
2010-2020	Total employment growth	-0.035	[-0.067, -0.001]	96

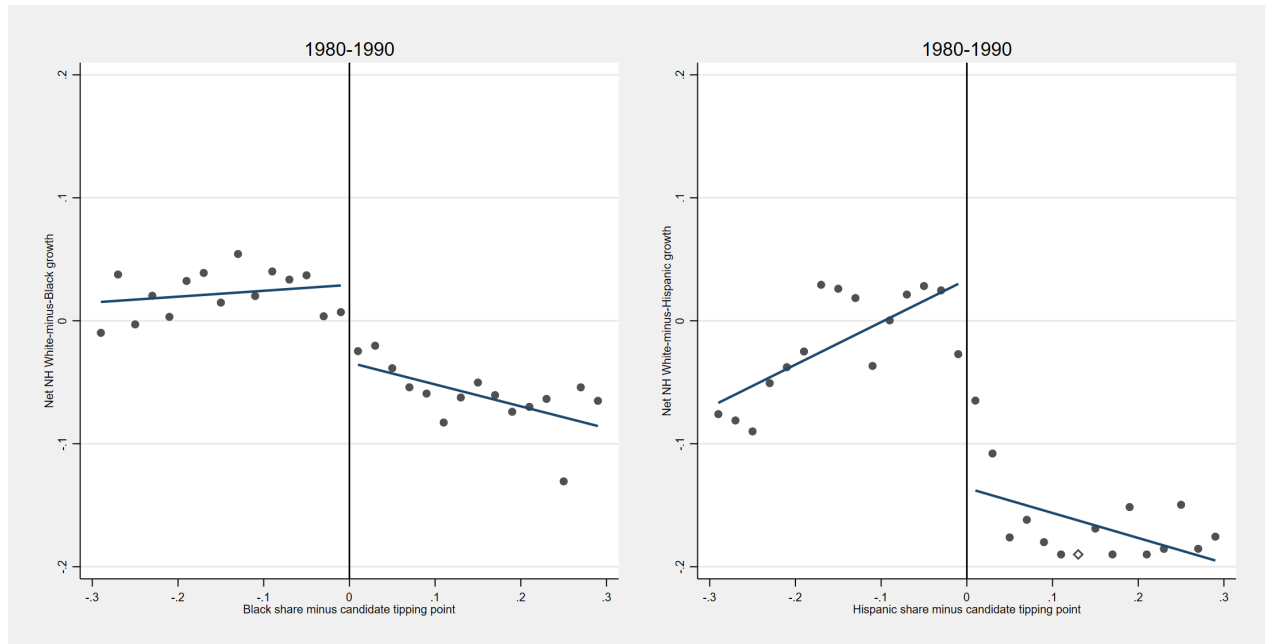
Notes: Candidate points are selected from non-Hispanic White employment growth in the same two-thirds search sample and are held fixed across rows, so each period compares outcomes at one location. Other details follow Table 4.

A.2 An Alternative Structural-Break Design

The fixed-point procedure locates the minority share at which predicted White employment growth equals the group average. An alternative is the structural break method, searching directly for the share at which the outcome falls most sharply (Card et al. 2008; Pan 2015). This appendix reports that design, run on the same cells and the same 500 partitions, with two differences. The outcome is net White-minus-minority employment growth rather than White growth alone following Pan (2015), and a group contributes a candidate point only when its first-stage break has a t -statistic below -1.96 and its search sample holds at least 40 cells. The screen is binding. Between 5.9 and 14.1 of the eighteen groups survive it, against 16.5 to 17.7 under the fixed-point procedure, so fewer cells enter the estimation sample for the second step.

The two designs agree on which episode is strongest. Appendix Table A5 shows a 1980–1990 Hispanic estimates with the mean of -0.067 , negative in 96 percent of partitions and significant in 43 percent. Every other estimated discontinuity in this table is smaller, and has a range that covers zero. Appendix Figure A5 illustrates the second-step estimates for both minority groups in 1980–1990. Because the outcome is relative growth rather than White growth, these estimates confirm the overall pattern observed in Table 3 rather than replicating its exact magnitudes.

Figure A5: Black-share and Hispanic-share discontinuities, 1980–1990



Notes: The left panel uses the Black employment share and net non-Hispanic White-minus-Black employment growth. The right panel uses the Hispanic share and net non-Hispanic White-minus-Hispanic growth. Dots are two-percentage-point bin means averaged over 500 repeated 2/3–1/3 partitions; contributing cells are always in the estimation sample. Lines are fitted separately on each side.

Table A5: Split-sample estimates of racial and ethnic occupational tipping

Period	Repeated-split mean	5th–95th percentile	Share $\hat{d} < 0$	Share $\hat{d} < 0$, split $p < .05$	Mean second- stage cells	Mean groups retained
<i>Panel A: Black employment share</i>						
1980-1990	-0.040	[-0.09, 0.02]	89	17	1510	7.9
1990-2000	-0.015	[-0.06, 0.04]	72	4	1469	6.8
2000-2010	-0.026	[-0.06, 0.01]	87	14	1296	5.9
2010-2020	-0.025	[-0.06, 0.01]	90	19	1920	9.1
<i>Panel B: Hispanic employment share</i>						
1980-1990	-0.067	[-0.12, -0.01]	96	43	1623	9.9
1990-2000	-0.022	[-0.07, 0.03]	78	12	1890	11.0
2000-2010	-0.007	[-0.03, 0.02]	68	5	2785	14.1
2010-2020	-0.009	[-0.04, 0.02]	69	4	2622	13.0

Notes: The analysis uses the split-sample logic of Card et al. (2008). Within each panel, the outcome is the decadal change in non-Hispanic White employment minus the change in employment of the stated group, divided by base-year total employment; the running variable is that group's base-year share. Black workers include those of Hispanic origin, and Hispanic origin is defined independently of race. Within each partition, the candidate tipping point is estimated on two-thirds of the sample and the discontinuity on the remaining one-third; the fold roles are not swapped. The table summarizes 500 independently generated partitions. The fifth-to-ninety-fifth percentile range and the two shares are split-stability diagnostics, not confidence intervals or formal tests of the repeated-split mean. Per-partition standard errors are clustered by state. The unit of observation is an occupation-state cell with at least 30 observations in both years. Candidate tipping points are retained for groups with at least 40 observations and a break t -statistic below -1.96 .